

NLP for Teens

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■ Session 1 Review — NLP Basics & Hugging Face

1. NLP is basically how we teach computers to understand the way we talk and write
2. Hugging Face is a huge library of ready NLP and AI models — you can use them instantly without training anything.

■ Session 2 Review — NLP Pipeline & Lemmatization

The NLP pipeline is just like any pipeline:

You put text in, do a few steps, and get results out.

1) Data acquisition

Getting the raw text you want to work with.

2) Cleanup

Removing unwanted or noisy text.

3) Tokenization

Breaking text into smaller pieces (words, subwords).

4) Vectorization

Turning words into numbers so models can understand them.

■ Session 2 Review — NLP Pipeline & Lemmatization

Stemming vs Lemmatization

Both try to make words simpler, but they do it in different ways.

- Stemming cuts off the end of a word. The result is sometimes not a real word.

running → runn

- Lemmatization finds the real dictionary word.

running → run

In short, stemming is faster but less accurate, while lemmatization is more accurate but a bit slower.

■ Session 3 Review — POS Tagging, NER & Bag-of-Words

POS = Part-of-Speech

grammar role

noun, verb, adjective...

NER = Named Entity Recognition

real-world label

person, country, company...

BoW = count words

CountVectorizer = BoW tool

- BoW = counts how many times each word appears

- CountVectorizer = BoW implementation

■ Session 4 Review—POS •NER•BoW •Stopwords •N-gra

POS

grammatical role of each word

NER

real-world label of each word

BOW

count how many times each word appears

Stopwords = common words that add little meaning.

Before:

The food is very good and the service is nice.

After (stopwords removed):

food, good, service, nice

■ Session 4 Review—POS •NER•BoW •Stopwords •N-gram

n-gram means a group of n words together.

- unigram → 1 word → food
- bigram → 2 words → food is
- trigram → 3 words → food is good

n-grams show word patterns, not just single words.

■ Session 5 Review — TF-IDF • Spam Detection

tf-idf = finds important words by combining how often a word appears (tf) and how rare it is across all documents (idf).

tf-idf simply picks out the important words and ignores the common ones.

■ Session 6 Review — GenAI • BART • Encoder-Decoder

nlp → transformers → llms → genai

nlp creates basic text understanding → transformers add deep context → llms scale it massively → genai produces useful outputs.

transformers read all words at once and use attention to understand how they relate. self-attention lets every word look at every other word.

genai means creating new content across text, images, code, and audio/video.

BART = BERT-style encoder + GPT-style decoder

1. encoder_only → understanding

bert, roberta, distilbert

2. decoder_only → generation

gpt, llama

3. encoder_decoder → translation and summarization

t5, bart

■ Session 7 Review — Modern NLP Tokenizers

tokens are the small pieces of text an llm actually reads.

llms don't understand full words — they understand tokens.

llms don't read full words — they read tokens.

tokenization turns raw text into these small pieces so the model can process language.

bpe is a simple way for gpt to break words into smaller pieces.

start with letters → find frequent pairs → merge them → keep merging until you get useful subwords.

Assignment

- Assignment: Text-to-Vector Conversion (Blog Authorship Corpus)
- [Dataset](#)
 - 1. Use the Blog Authorship Corpus dataset on Kaggle.
 - 2. Take the text column.
 - 3. Convert the text into numerical vectors using any method from class.

Summary

- ✓ NLP basics review
- ✓ pipeline: data → cleanup → tokenization → vectorization
- ✓ stemming cuts words, lemmatization finds dictionary forms
- ✓ POS = grammar role, NER = real-world label, BoW = word counts
- ✓ stopwords remove common words, n-grams show word patterns
- ✓ tf-idf finds important words
- ✓ transformers use attention, encoder/decoder models
- ✓ BART = encoder + decoder
- ✓ tokens small text pieces LLMs read
- ✓ WordPiece “##”, BPE merges pairs, SentencePiece “_”

Thank You

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